MIL-D-3263B
20 July 1987
SUPERSEDING
MIL-D-3263A
5 August 1968
(See 6.6)

MILITARY SPECIFICATION

DESICCANT CONTAINERS, DEHUMIDIFIER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers desiccant containers and their storage cases which are suitable for the static dehumidification of small, enclosed spaces, and which can be readily reactivated.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4440
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SPECIFICATIONS

FEDERAL

QQ-A-200/8 - Aluminum Alloy 6061, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-250/11 - Aluminum Alloy 6061, Plate and Sheet.
QQ-S-698 - Steel, Sheet and Strip, Low-Carbon.
VV-S-190 - Sealing Compound (Dipcoat).
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

DOD-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-3716 - Desiccants, Activated for Dynamic Dehumidification.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
DOD-STD-100 - Engineering Drawing Practices.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 229 - Standard Specification for Steel Wire, Oil-Tempered for Mechanical Springs. (DoD adopted)
A 386 - Standard Specification for Zinc Coating (Hot-Dip) on Assembled Steel Products.
B 6 - Standard Specification for Zinc (Slab Zinc). (DoD adopted)
B 633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials used in the construction of the desiccant container and the storage case shall be free from defects, capable of being punched or formed, or both, without the creation of cracks and suitable for the purpose intended.

3.1.1 Recovered materials. Unless otherwise specified herein, all material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used products is allowed under this specification unless otherwise specifically specified.

3.2 Container. The container shall consist of two circular side plates, an end band (or ring), a handle, a receptacle for an indicating cartridge, built as a single unit, ready for installation.

3.2.1 Configuration. The container shall have contours, dimensions, and tolerances in accordance with figure 1.

3.2.2 Side plates and end band. The container side plates and end band (or ring) shall be made of aluminum in accordance with QQ-A-250/11. The side plates shall be at least 0.032 inch thick and each plate shall be uniformly perforated. The perforated area shall be not less than 27 percent of the total effective area of each plate with the perforation having a maximum diameter of 0.033 inch. The minimum number of perforations shall be 324 per square inch. The end band (or ring) shall be at least 0.050 inch thick. The side plates and end band shall be fastened together by spot welding, brazing, or by blind rivets.

3.2.2.1 Staybolts. Three 3/8 inch staybolts shall be fastened, uniformly spaced at 120-degree intervals and approximately 2-1/2 inches from the center of the container. The staybolts shall be of aluminum in accordance with QQ-A-200/8 and shall be fastened flush against the surface of the side plates.

3.2.3 Handle. Each container shall be provided with a handle. The handle shall be constructed of aluminum in accordance with QQ-A-250/11 and shall be at least 0.125 inch thick. The handle shall be provided with a hole not less than 3/4 inch in diameter for handling the desiccant container. The handle shall be securely fastened to the end band (or ring) by spot welding, brazing, or by blind rivets.

3.2.4 Receptacle. Each container shall be provided with a receptacle located in the end band (or ring) approximately 30 degrees from the handle,

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into which a 3/4-inch indicating cartridge can be screwed. The receptacle shall be of aluminum in accordance with QQ-A-250/11 and shall be perforated at the bottom or provided with a screen, to retain the desiccant and permit circulation of air between the desiccant and the indicating cartridge. The receptacle shall be either deep enough to permit insertion of the cartridge flush with the end band (or ring), or shall be provided with a collar to protect the indicating cartridge. The indicating cartridge shall be in accordance with figure 2, and shall be provided in the receptacle of each container (see 6.3).

3.2.5 Desiccant. Each container shall be filled with at least 5 pounds of activated desiccant conforming to grade H, type I or II of MIL-D-3716.

3.2.5.1 Indicator desiccant. The indicator cartridge shall be filled with activated desiccant conforming to grade H, type IV of MIL-D-3716.

3.3 Airtight storage case. The storage case shall consist of a cylindrical body with two drop-type handles, bottom panel, a cover (see 3.3.6), with a handle and four trunk-type clamps. When subjected to an internal air pressure of 2 inches of water, the pressure drop shall not exceed 0.2 inch of water in 10 minutes. The case shall satisfactorily accommodate eight of the desiccant containers specified in 3.2.

3.3.1 Construction. The configuration, physical dimensions and tolerances of the storage case shall be in accordance with figure 3.

3.3.2 Body. The body of the storage case shall be made of black steel in accordance with QQ-S-698 and shall be at least 0.034 inch thick. The body shall have at least three horizontal strengthening beads. The side joints shall be lock seamed, welded or secured by blind rivets. The top edge shall be rolled over a 3/16 inch diameter steel wire to form a tightly closed head.

3.3.3 Bottom panel. The bottom panel shall be flat and formed in one piece from black steel in accordance with QQ-S-698. The panel shall be at least 0.045 inch thick, and shall be joined to the body by a double seam or seam weld.

3.3.4 Handles. The storage case shall be provided with two identical drop-type handles. The handles shall be made of black steel wire in accordance with ASTM A 229. The wire shall be at least 5/16 inch in diameter. The handles shall be attached to the body by means of handle clips formed from black steel in accordance with QQ-S-698. The clips shall be at least 0.045 inch thick and shall be located over the side seams (see figure 3), diametrically opposite each other. The clips shall be securely fastened to the body with at least four spot welds. The clips shall be provided with stops so the handle cannot be elevated through an angle greater than 80 degrees.

3.3.5 Clamps. Each storage case shall be provided with four trunk-type clamps uniformly spaced around the top edge of the body and 45 degrees from the drop handles. The clamps shall consist of a bracket to which shall be riveted a handle fitted with a steel wire claw. The bracket and handle shall be made of black steel in accordance with QQ-S-698 and shall be at least 0.045 inch thick. The claw shall be made of steel wire in accordance with ASTM A 229 and shall be at least 0.135 inch in diameter. The bracket shall fit tightly

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against the top bead and shall be securely fastened to the body by at least two spot welds or by at least two blind rivets. Each clamp shall be provided with a hole approximately 1/8 inch in diameter to receive a sealing wire (see figure 3).

3.3.6 Cover. Each storage case shall be provided with a snug fit cover. The cover shall be made of black steel in accordance with QQ-S-698 and shall be at least 0.045 inch thick. The cover shall be single piece construction, with the cover edges formed into a semicircular flange to fit tightly over the beaded edge of the body. The cover shall be concave and recessed at least 1/2 inch into the body.

3.3.6.1 Cover handle. A folding type handle shall be provided on the top side of the cover and shall not interfere with stowing sealed cases one on top of each other. The handle shall not project beyond the top edges of the semicircular flange. The handle shall be made of steel wire in accordance with ASTM A 229 and shall be attached to the center of the cover by a steel clip. The wire shall be at least 5/16 inch in diameter. The clip shall be at least 0.045 inch thick. The clip shall be securely fastened to the cover with at least four spot welds or by four blind rivets.

3.3.7 Finish. After fabrication, the storage case, including the support brackets of the clamps and the handle clip, shall be coated both inside and out with zinc by the hot-dip process, or electroplated. The handle and wire spring to the clamps shall be electroplated with zinc. The hot-dipped zinc coating shall be in accordance with ASTM A 386 with the spelter conforming to ASTM B 6. The electroplated zinc coating shall conform to ASTM B 633.

3.4 Identification plate. Each desiccant container and each storage case shall be provided with a single identification plate. The type of marking shall be Gothic capitals for letters, and numerals, and other characters shall be of similar appearance. The size of the letters, numerals and other characters shall be as necessary for easy readability except for the storage instructions on the storage case cover which shall be as specified in 3.4.3. The data applicable to the desiccant container shall be stamped or embossed into the handle of the container. The data applicable to the storage case shall be stamped or embossed into the cover of the storage case. The markings on the storage case cover shall be filled with black enamel to insure legibility.

3.4.1 Identification plate information. The identification plate shall contain the following information:

(a) Item name.
(b) Federal stock number.
(c) Weight (see 3.4.1.1), as applicable.
(d) Contract number.
(e) Manufacturer's name and address.

3.4.1.1 Weight information. Each completed desiccant container shall be weighed with the desiccant in the reactivated condition and the gross weight to the nearest tenth of a pound shall be added to the identification plate.

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3.4.2 Instructions (desiccant container only). The instructions for reactivation of the desiccant shall be as shown on figure 1.

3.4.3 Instructions (storage container only). The storage instructions shall be as shown on figure 3. Instruction markings shall be at least 1/2 inch in height.

3.5 Drawings. Drawings delineating the equipment shall be furnished in accordance with DOD-D-1000 and DOD-STD-100.

3.6 Workmanship. Desiccant containers and storage cases shall be free from defects that effect their appearance or operation. The container end band and the storage case body and cover shall show no cracks due to forming. Sharp edges, burrs and imperfections shall be removed from all parts subject to contact with personnel. The hot-dipped coatings shall be smooth, free from uncoated spots, injurious lumps, blisters and dross or flux.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. For the purpose of sampling and inspection, a lot shall consist of not more than 1000 storage cases, complete with containers, offered for delivery at the same time.

4.2.2 Sampling for examination. Sample storage cases, complete with containers, shall be selected at random from each lot in accordance with table I for the examination specified in 4.3.

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TABLE I. Sampling.

Lot size no. of cases	Sample size no. of cases	Acceptance number (defectives)	Rejection number (defectives)
1 to 8	All	-	-
9 to 15	5	0	1
16 to 25	7	0	1
26 to 40	10	0	1
41 to 65	15	0	1
66 to 110	15	0	1
111 to 180	25	1	2
181 to 300	25	1	2
301 and over	35	2	3

4.2.3 Sampling for leakage test. Sample storage cases shall be selected at random from each lot for the test specified in 4.4. A sample shall constitute 40 percent of each lot sample that is selected at random in accordance with table I for the examination specified in 4.3. Any storage case in the sample in which there is evidence of excessive leakage when tested shall not be offered for delivery, and if the number of defective cases in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall not be offered for delivery.

4.3 Examination. Each of the sample storage cases (including containers) selected in accordance with 4.2.2 shall be examined and measured to verify conformance to the requirements of this specification not involving tests. Examination shall be conducted as specified in table II. Any storage case or desiccant container in the sample containing one or more defects shall not be offered for delivery, and if the number of defective units (storage case and containers) in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall not be offered for delivery.

TABLE II. Classification of defects.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Materials defective or not as specified.
102	Drawings not followed.
103	Overall dimensions not within specified tolerances.
104	Handles missing, damaged, or not as specified.
105	Storage case bottom panel not single piece construction or not as specified.
106	Storage case body not constructed as specified.
107	Storage case cover not single piece construction, damaged, missing, or not as specified.

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TABLE II. Classification of defects. - Continued

Categories	Defects
108	Zinc coating damaged, incomplete, or not as specified.
109	Desiccant containers damaged or missing.
110	Desiccant not in container or underweight.
111	Indicator cartridge damaged, missing, or not as specified.
112	Container end band (or ring) damaged or not as specified.
113	Container side plates not perforated, damaged, or not as specified.
114	Container staybolts missing or not as specified.
115	Receptacle for indicating cartridge not furnished or as specified.
116	Welding or brazing not as specified.
117	Identification plate marking not permanent, illegible or not as specified.
118	Instruction plate data missing, not permanent, illegible, or not as specified.
Minor:	
201	Storage case not sealed or sealing wire missing (see 5.1).
202	Sharp edges, burrs and imperfections on parts.
203	Coatings not smooth, uncoated spots, injurious lumps, blisters and dross or flux.
204	Level of preservation, packaging and packing not as specified (see 5.1 and 5.2).
205	Boxes not type or class specified.
206	Markings on interior and exterior units omitted, incorrect or illegible.

4.4 Leakage test for complete storage case. Each sample (see 4.2.3) storage case shall be subjected to an internal air pressure of 2 inches of water. A pressure drop greater than 0.2 inch of water over a period of 10 minutes shall constitute failure of this test.

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

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5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each storage case shall be sealed watertight with eight desiccant containers in the reactivated condition and as follows:

(a) The storage case cover edges shall be sealed watertight with sealing compound conforming to VV-S-190.
(b) The cover clamps of each storage case shall be secured to prevent opening during shipment (see 3.3.5).

5.1.2 Level C. Desiccant containers shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each complete unit, preserved and packaged as specified in 5.1.1, shall be packed one each to a shipping container. Containers shall conform to class II of PPP-B-591, overseas type of PPP-B-601, class 2 of PPP-B-621, or weather resistant class of PPP-B-636. Each fiberboard shipping container shall be waterproofed by means of tape in accordance with the appendix of the container specification. Wood or cleated-plywood containers shall be provided with a type I or II, grade B caseliner conforming to MIL-L-10547. Each shipping container shall be closed, reinforced with flat strapping or tape banding in accordance with the appendix of the applicable container specification.

5.2.2 Level B. Each complete unit, preserved and packaged as specified in 5.1.1, shall be packed one each to a shipping container. Containers shall conform to class I of PPP-B-591, domestic type of PPP-B-601, class I of PPP-B-621, or domestic class of PPP-B-636. Shipping containers shall be closed in accordance with applicable box specification.

5.2.3 Level C. Each complete unit, preserved and packaged as specified in 5.1.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The desiccant containers covered by this specification are intended for use with readily reactivated desiccant for use in small enclosed spaces.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable preservation, packaging, packing and marking (see 5.1 and 5.2).

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6.3 Indicating cartridge. The indicating cartridge specified in 3.2.4 is a Navy standard humiplug, Federal Stock Number H6685-244-1029.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Container
Dehumidification
Desiccant
Reactivation

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

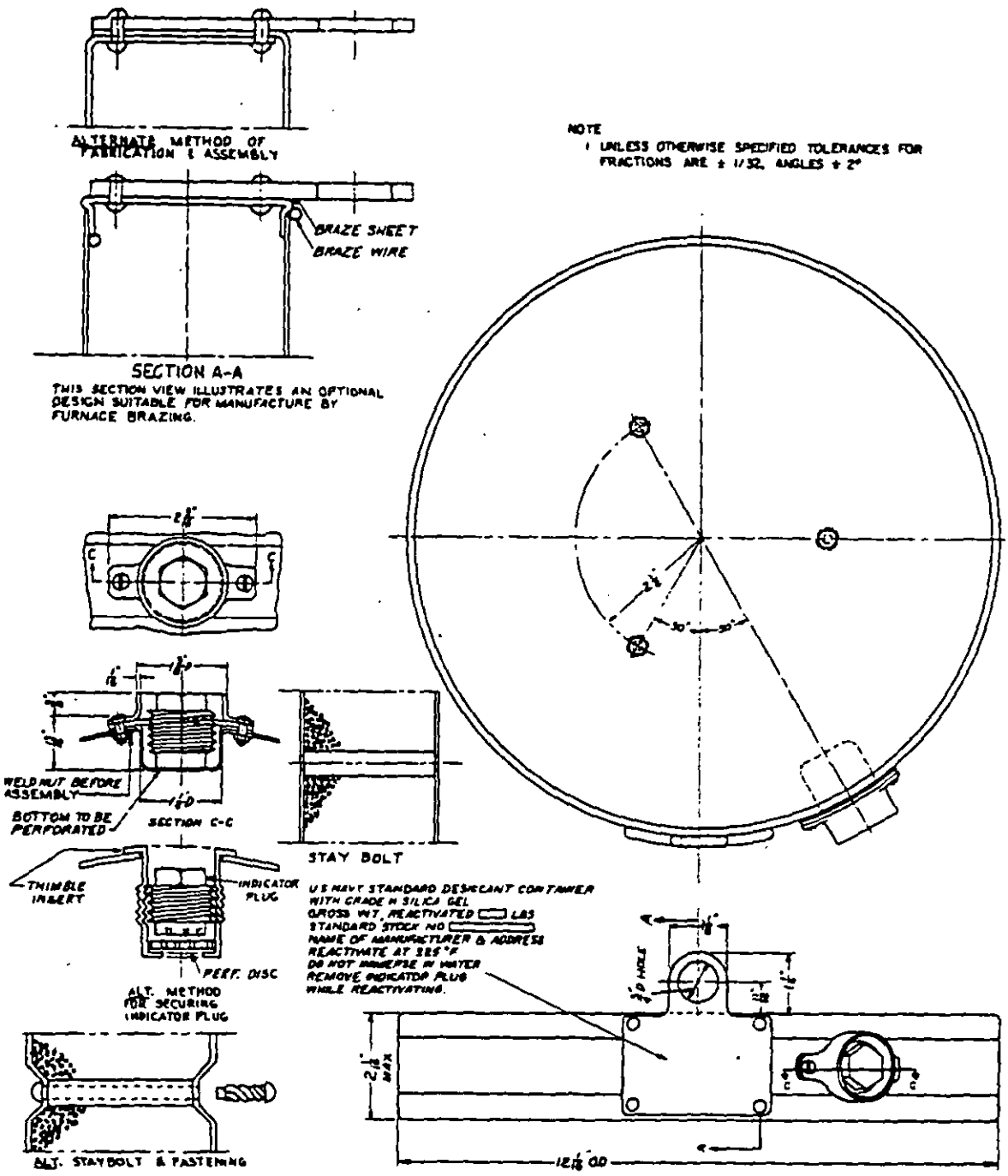
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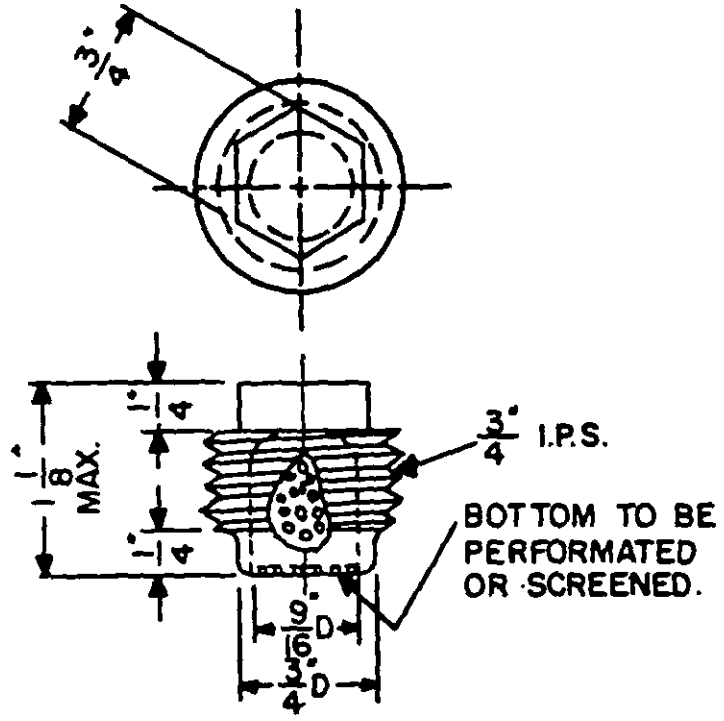
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SH 3319A

FIGURE 1. Desiccant container.

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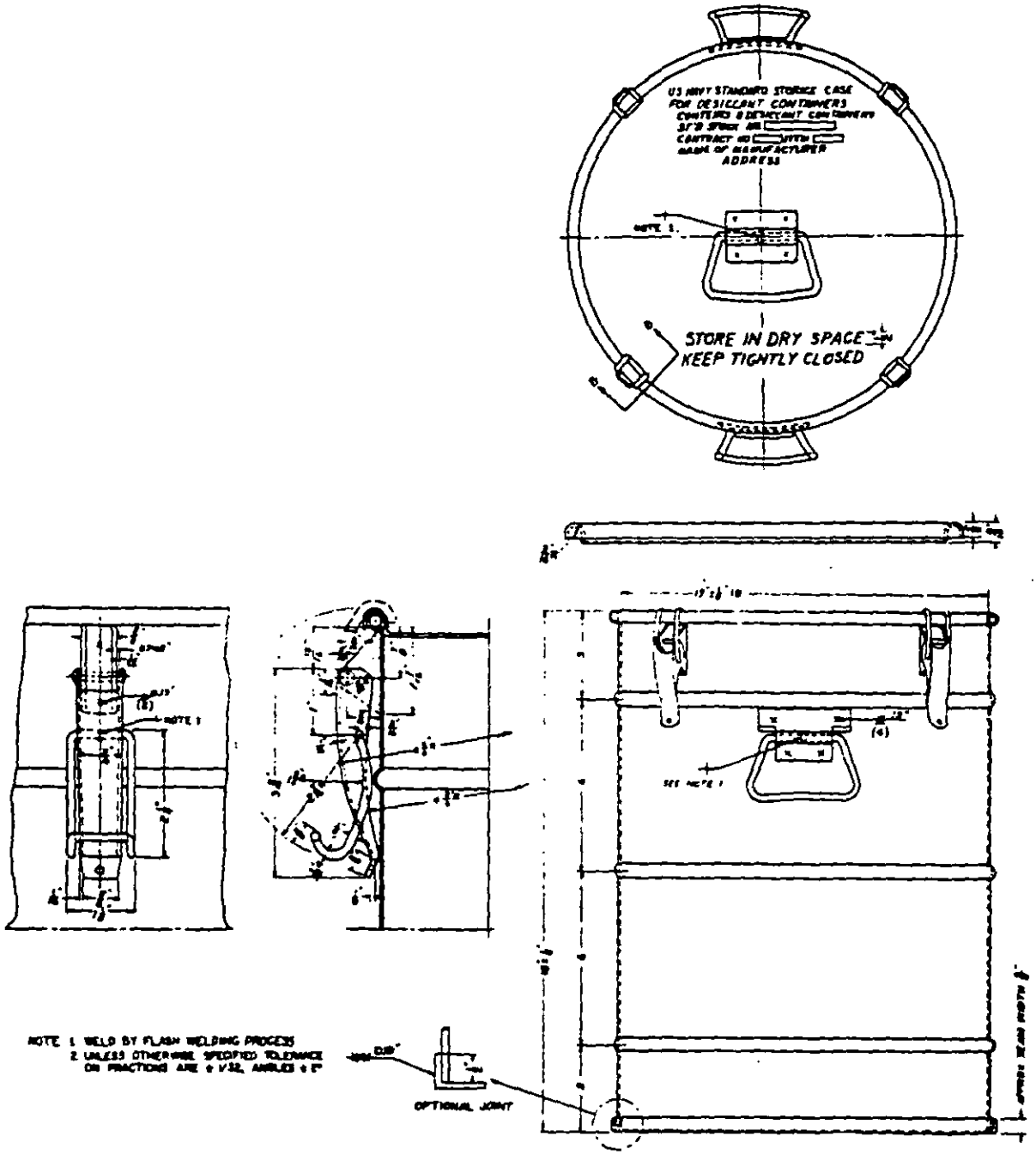
NOTE

1. UNLESS OTHERWISE SPECIFIED TOLERANCES FOR FRACTIONS ARE ± 1/64.
2. INDICATOR CARTRIDGE SHALL BE MADE OF CLEAR PLASTIC OR THE TOP CONSTRUCTED SO INDICATOR CAN BE READ WITHOUT REMOVING FROM RECEPTACLE.

SH 9158

FIGURE 2. Indicator cartridge.

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SH 3320A

FIGURE 3. Storage case.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-3263B	DESICCANT CONTAINERS, DEHUMIDIFIER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	8. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	9. DATE OF SUBMISSION *(YYMMDD)*

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